

1. Создала две таблицы **level** и **user**.

```
CREATE TABLE level (  
  id serial not null,  
  level_name varchar(40) not null,  
  PRIMARY KEY (id)  
);
```

```
CREATE TABLE user (  
  id serial not null,  
  user_name varchar(30) not null,  
  level_id int not null,  
  skill int not null,  
  PRIMARY KEY (id),  
  FOREIGN KEY (level_id) REFERENCES  
  level(id)  
);
```

```
CREATE TABLE level (  
  id serial not null,  
  level_name varchar(40) not null,  
  PRIMARY KEY (id)  
);  
  
CREATE TABLE user (  
  id serial not null,  
  user_name varchar(30) not null,  
  level_id int not null,  
  skill int not null,  
  PRIMARY KEY (id),  
  FOREIGN KEY (level_id) REFERENCES level(id)  
);
```

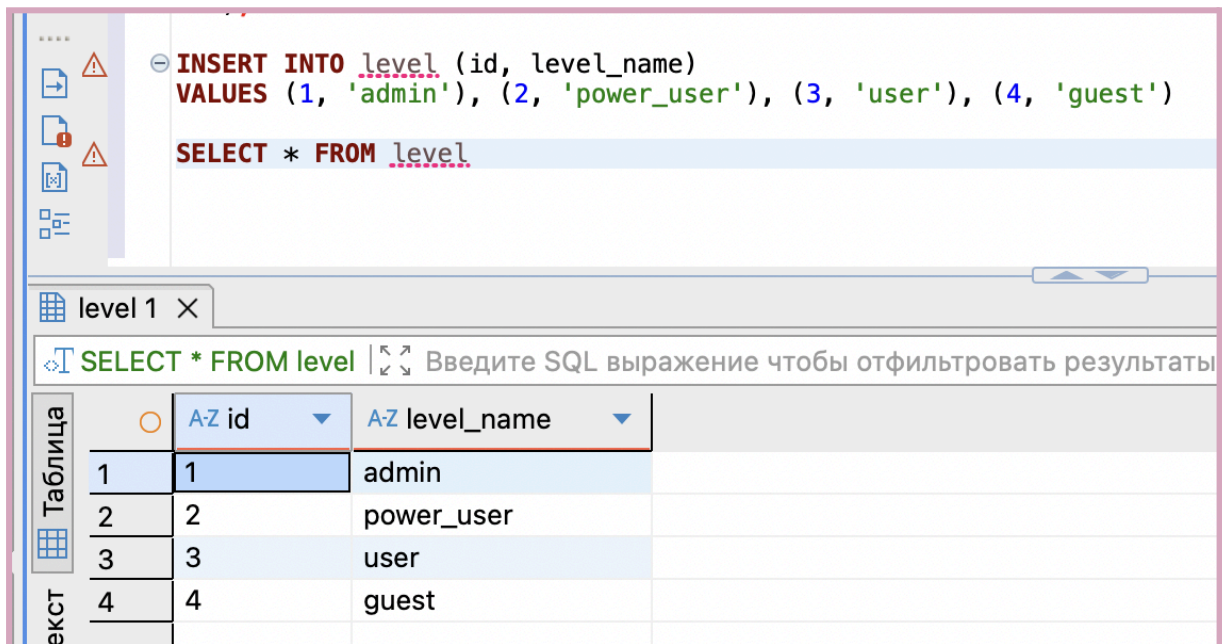
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LECT * FROM user | Введите SQL выражение чтобы отфильтровать результаты

	A-Z id	A-Z user_name	123 level_id	123 skill	

2. Внесла данные в таблицу level

```
INSERT INTO level (id, level_name)
VALUES (1, 'admin'), (2, 'power_user'), (3, 'user'), (4, 'guest')
```



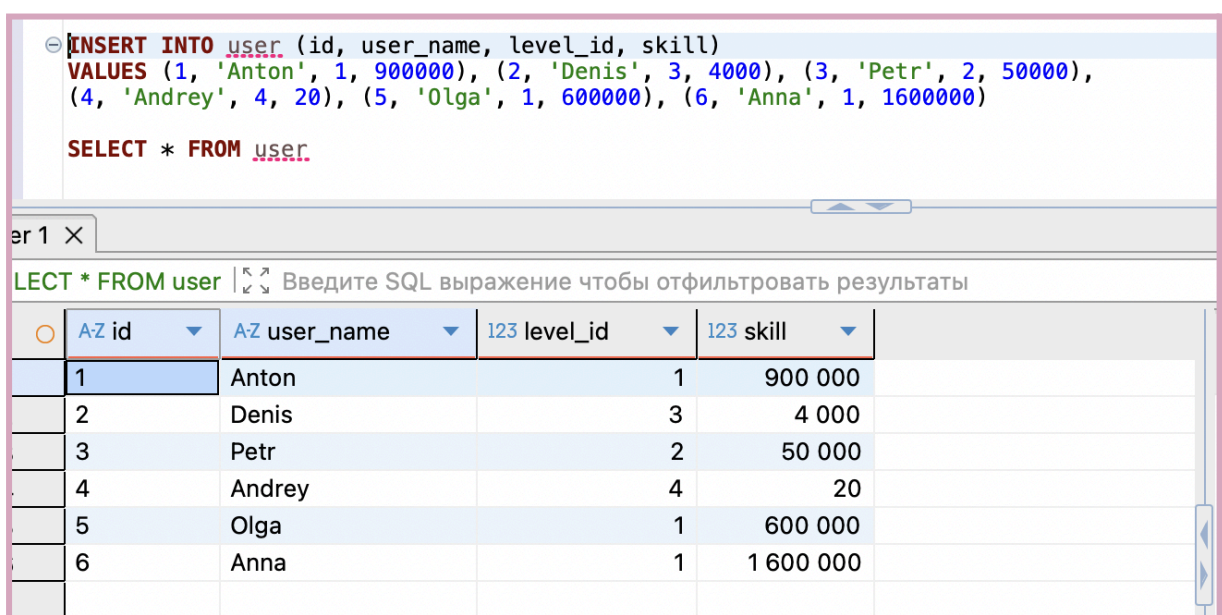
```
INSERT INTO level (id, level_name)
VALUES (1, 'admin'), (2, 'power_user'), (3, 'user'), (4, 'guest')

SELECT * FROM level
```

	A-Z id	A-Z level_name
1	1	admin
2	2	power_user
3	3	user
4	4	guest

3. Внесла данные в таблицу user

```
INSERT INTO user (id, user_name, level_id, skill)
VALUES (1, 'Anton', 1, 900000), (2, 'Denis', 3, 4000), (3, 'Petr', 2, 50000),
(4, 'Andrey', 4, 20), (5, 'Olga', 1, 600000), (6, 'Anna', 1, 1600000)
```



```
INSERT INTO user (id, user_name, level_id, skill)
VALUES (1, 'Anton', 1, 900000), (2, 'Denis', 3, 4000), (3, 'Petr', 2, 50000),
(4, 'Andrey', 4, 20), (5, 'Olga', 1, 600000), (6, 'Anna', 1, 1600000)

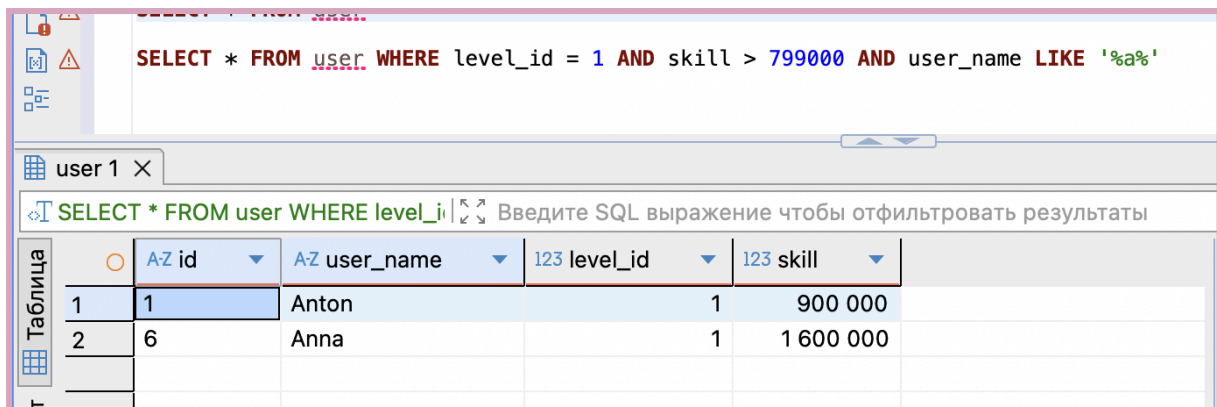
SELECT * FROM user
```

	A-Z id	A-Z user_name	123 level_id	123 skill
1	1	Anton	1	900 000
2	2	Denis	3	4 000
3	3	Petr	2	50 000
4	4	Andrey	4	20
5	5	Olga	1	600 000
6	6	Anna	1	1 600 000

Задания на написание запросов к БД:

1. Отобрать из таблицы user всех пользователей, у которых level_id=1, skill > 799000 и в имени встречается буква а

```
SELECT * FROM user WHERE level_id = 1 AND skill > 799000 AND user_name LIKE '%a%'
```

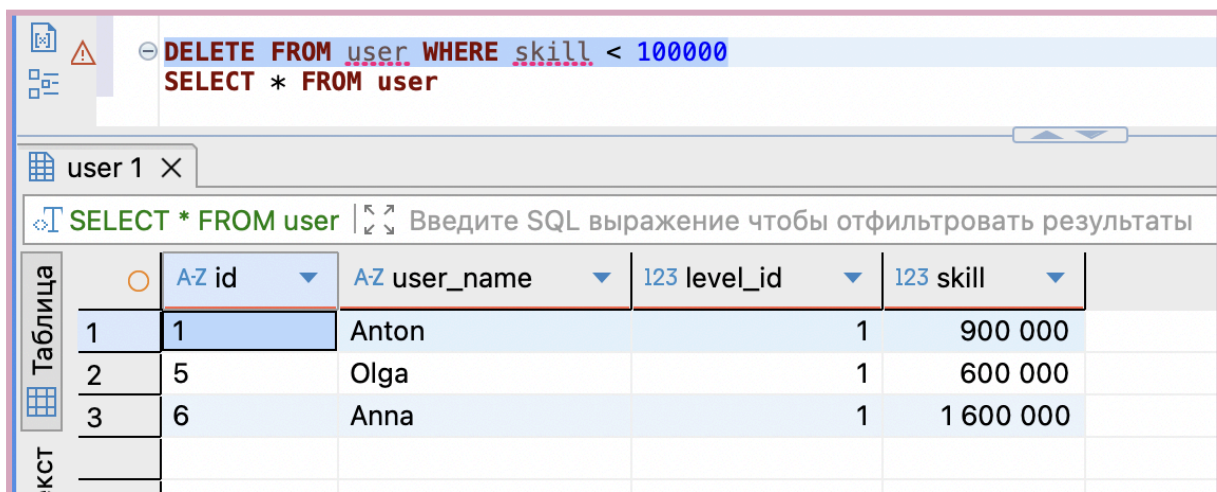


The screenshot shows a database query editor with the following SQL query: `SELECT * FROM user WHERE level_id = 1 AND skill > 799000 AND user_name LIKE '%a%'`. Below the query, a table titled "user 1" displays the results. The table has columns: id, user_name, level_id, and skill. The results show two rows: Anton (id=1, level_id=1, skill=900 000) and Anna (id=6, level_id=1, skill=1 600 000).

	A-Z id	A-Z user_name	123 level_id	123 skill
1	1	Anton	1	900 000
2	6	Anna	1	1 600 000

2. Удалить всех пользователей, у которых skill меньше 100000

```
DELETE FROM user WHERE skill < 100000
```

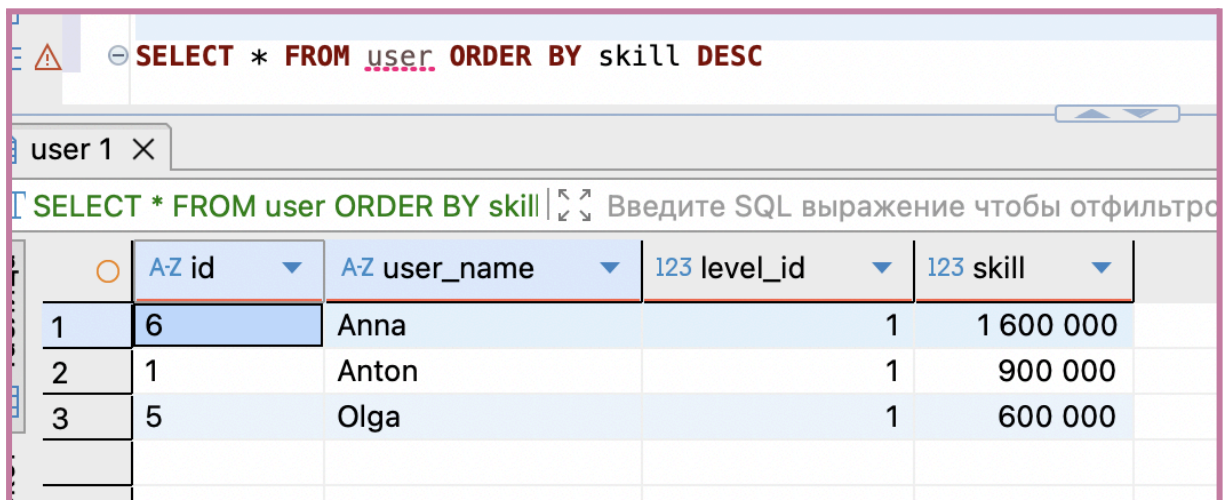


The screenshot shows a database query editor with the following SQL query: `DELETE FROM user WHERE skill < 100000`. Below the query, a table titled "user 1" displays the results. The table has columns: id, user_name, level_id, and skill. The results show three rows: Anton (id=1, level_id=1, skill=900 000), Olga (id=5, level_id=1, skill=600 000), and Anna (id=6, level_id=1, skill=1 600 000).

	A-Z id	A-Z user_name	123 level_id	123 skill
1	1	Anton	1	900 000
2	5	Olga	1	600 000
3	6	Anna	1	1 600 000

3. Вывести все данные из таблицы user в порядке убывания по полю skill

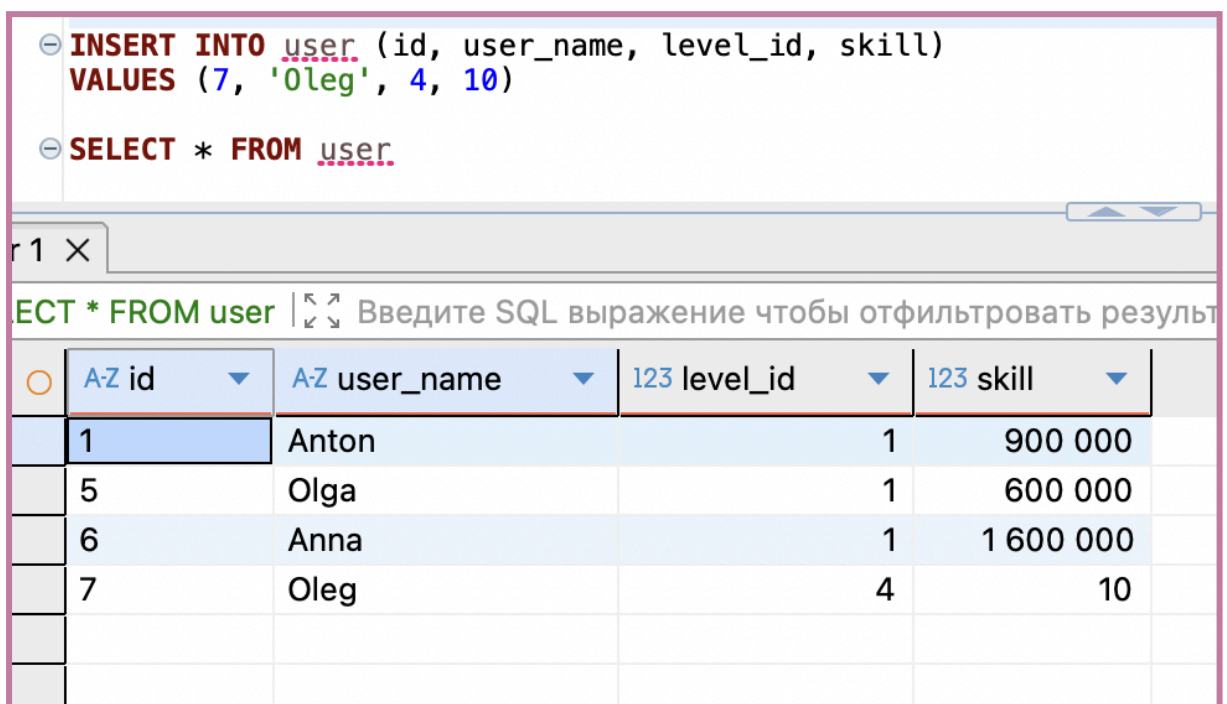
```
SELECT * FROM user ORDER BY skill DESC
```



	A-Z id	A-Z user_name	123 level_id	123 skill
1	6	Anna	1	1 600 000
2	1	Anton	1	900 000
3	5	Olga	1	600 000

4. Добавить в таблицу user нового пользователя по имени Oleg, с уровнем 4 и skill =10

```
INSERT INTO user (id, user_name, level_id, skill)  
VALUES (7, 'Oleg', 4, 10)
```



```
INSERT INTO user (id, user_name, level_id, skill)  
VALUES (7, 'Oleg', 4, 10)
```

```
SELECT * FROM user
```

	A-Z id	A-Z user_name	123 level_id	123 skill
	1	Anton	1	900 000
	5	Olga	1	600 000
	6	Anna	1	1 600 000
	7	Oleg	4	10

5. Обновить данные в таблице user - для пользователей с level_id меньше 2 проставить skill 2000000

```
UPDATE user SET skill = 2000000 WHERE level_id < 2
```

The screenshot shows a database client interface. At the top, the SQL statement `UPDATE user SET skill = 2000000 WHERE level_id < 2` is entered. Below it, the `SELECT * FROM user` statement is shown. The results are displayed in a table with columns: id, user_name, level_id, and skill.

	A-Z id	A-Z user_name	level_id	skill
1	1	Anton	1	2 000 000
2	5	Olga	1	2 000 000
3	6	Anna	1	2 000 000
4	7	Oleg	4	10

6. Выбрать user_name всех пользователей уровня admin используя подзапрос

```
SELECT user_name FROM user WHERE level_id IN (SELECT id FROM level WHERE level_name = 'admin')
```

The screenshot shows a database client interface. The SQL statement `SELECT user_name FROM user WHERE level_id IN (SELECT id FROM level WHERE level_name = 'admin')` is entered. The results are displayed in a table with columns: user_name.

A-Z user_name
Anton
Olga
Anna

7. Выбрать user_name всех пользователей уровня admin используя join

```
SELECT user_name FROM user u INNER JOIN level l  
ON u.level_id = l.id WHERE l.level_name = 'admin'
```

SELECT user_name FROM user u INNER JOIN level l
ON u.level_id = l.id WHERE l.level_name = 'admin'

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SELECT user_name FROM user u INNER JOIN level l
ON u.level_id = l.id WHERE l.level_name = 'admin'

Введите SQL выражение

A-Z user_name

Anton

Olga

Anna